



Diesel Generator Set

mtu 20V4000 DS3100

380V – 11 kV/50 Hz/standby power/NEA (ORDE) optimized
20V4000G74F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 3080 kVA - 3180 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	390
Model	20V4000G74F		Engine jacket water capacity: l	205
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	20V		Combustion air requirements	
Displacement: l	95.4			
Bore: mm	170			
Stroke: mm	210		Combustion air volume: m³/s	2.6
Compression ratio	16.4		Max. air intake restriction: mbar	50
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ECU 9			
Max power: kWm	2670			
Air cleaner	dry		Coolant flow rate (HT circuit): m³/hr	80
Fuel system			Coolant flow rate (LT circuit): m³/hr	32.5
			Heat rejection to coolant: kW	1030
			Heat radiated to charge air cooling: kW	490
			Heat radiated to ambient: kW	105
			Fan power for electr. radiator (40°C): kW	70
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	27			
Fuel consumption ²⁾				
	l/hr	g/kwh	Exhaust gas temp. (after turbocharger): °C	550
At 100% of power rating:	633.7	197	Exhaust gas volume: m³/s	8.6
At 75% of power rating:	494.6	205	Maximum allowable back pressure: mbar	85
At 50% of power rating:	345.8	215	Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 M9 (Low voltage Leroy Somer standard)	380 V	2512	3140	4771	2472	3090	4695
	400 V	2512	3140	4532	2472	3090	4460
	415 V	2512	3140	4368	2472	3090	4299
Marathon 1030FDL7094 (Low voltage Marathon)	380 V	2536	3170	4816	2464	3080	4680
	400 V	2536	3170	4576	2464	3080	4446
	415 V	2536	3170	4410	2464	3080	4285
Marathon 1030FDH7101 (Medium volt. marathon)	11 kV	2536	3170	166	2472	3090	162
Leroy Somer LSA53.2 ZL14 (Medium volt. Leroy Somer)	11 kV	2544	3180	167	2472	3090	162

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.